

RUSHIT KHAKHKHAR

B.Tech Student — Information Technology | Vellore Institute of Technology, Vellore

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PROFESSIONAL SUMMARY

Information Technology undergraduate at VIT Vellore with practical expertise in Machine Learning, Big Data Engineering, and Full-Stack Web Development. Proven experience architecting and deploying end-to-end data pipelines processing millions of records using Python, Apache Spark, and Scikit-learn. Skilled in building interactive Streamlit applications deployed on cloud platforms and developing MERN stack applications. Strong foundation in Data Structures, Algorithms, DBMS, and OOP.

EDUCATION

Vellore Institute of Technology — B.Tech in Information Technology Vellore, Tamil Nadu CGPA: 8.0	Aug 2023 – Present (Expected 2027)
Cosmos Castle International School — CBSE Class XII (Science PCM) Ahmedabad, Gujarat Score: 72%	2023
Ambuja Vidya Niketan — CBSE Class X Ambujanagar, Gujarat Score: 92%	2021

TECHNICAL SKILLS

Languages:	Java, Python, C, C++, JavaScript, R, MATLAB, SQL
Data Science & ML:	Scikit-learn, TensorFlow, OpenCV, XGBoost, Random Forest, ARIMA, Pandas, NumPy, Matplotlib, Plotly, Streamlit
Big Data & DB:	Apache Spark (PySpark), ETL Pipelines, MySQL / Oracle SQL, MongoDB
Web Development:	HTML5, CSS3, JavaScript, React.js, Node.js, Express.js, REST APIs (MERN Stack)
Tools & Platforms:	Git, GitHub, VS Code, Jupyter Notebook, Tableau, Hugging Face Spaces, Streamlit Cloud
Core Concepts:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, SDLC

PROJECTS

Smart City Water Leakage Detection System • Engineered an Apache Spark big data pipeline analyzing 4.32 million+ smart meter records; developed spike-ratio anomaly detection classifying severity from Normal to Critical. • Trained Random Forest and XGBoost classifiers on 10 behavioral features; deployed an 11-page interactive dashboard on Hugging Face Spaces and Streamlit Cloud with What-If simulations.	<i>Python, Apache Spark, Random Forest, XGBoost, Streamlit, Plotly</i>
Crime Intelligence & Predictive Forecasting System • Processed ~1.45 million Chicago crime records, using KMeans clustering for High/Medium/Low spatial risk categorization and centroid hotspot mapping. • Implemented ARIMA time-series forecasting and optimized a Random Forest regressor (RMSE: 48.31) using lag and rolling features within an interactive multi-page Streamlit dashboard.	<i>Python, Scikit-learn, XGBoost, ARIMA, KMeans, Streamlit</i>
Facial Expression Detection Using Deep Learning • Constructed a deep learning pipeline utilizing OpenCV for facial feature extraction and region-of-interest preprocessing from image datasets. • Architected and trained a custom Multi-Layer Perceptron (MLP) in TensorFlow, achieving robust multi-class facial expression classification.	<i>Python, TensorFlow, OpenCV, NumPy, Pandas</i>
Kaithi Ayurveda — E-Commerce Web Application • Developed a modular Ayurveda e-commerce web platform in Streamlit featuring dynamic search/filter, custom CSS styling, and responsive UI components. • Integrated an interactive in-browser 3D product visualization viewer using procedural Three.js geometry with GLB/GLTF model support.	<i>Python, Streamlit, HTML/CSS, JavaScript, Three.js</i>

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — Oracle | Aug 2025 | ID: 102415557OCI25GAIOCP
- **Oracle Fusion AI Agent Studio Foundations Associate – Rel 1** — Oracle | Aug 2025 | ID: 102415557OFAASOFA
- **Introduction to Generative AI** — Simplilearn | Aug 2025 | Code: 8849908
- **Certificate of Participation — Yantra Central Hackathon 2025** — Office of Students' Welfare, VIT Vellore | Feb 2025

ACHIEVEMENTS & ACTIVITIES

- **Volunteer, Gravitas Tech Fest, VIT** — Assisted in organizing and coordinating tech-related events during the institute's annual technical festival.
- **Organizer, Genesis Hackathon on Game Development** — Led the planning and execution of a game development-focused hackathon, managing logistics, outreach, and event flow.
- Deployed 2 scalable Machine Learning applications publicly on **Hugging Face Spaces** and Streamlit Cloud, processing 1M+ real records via Apache Spark.